



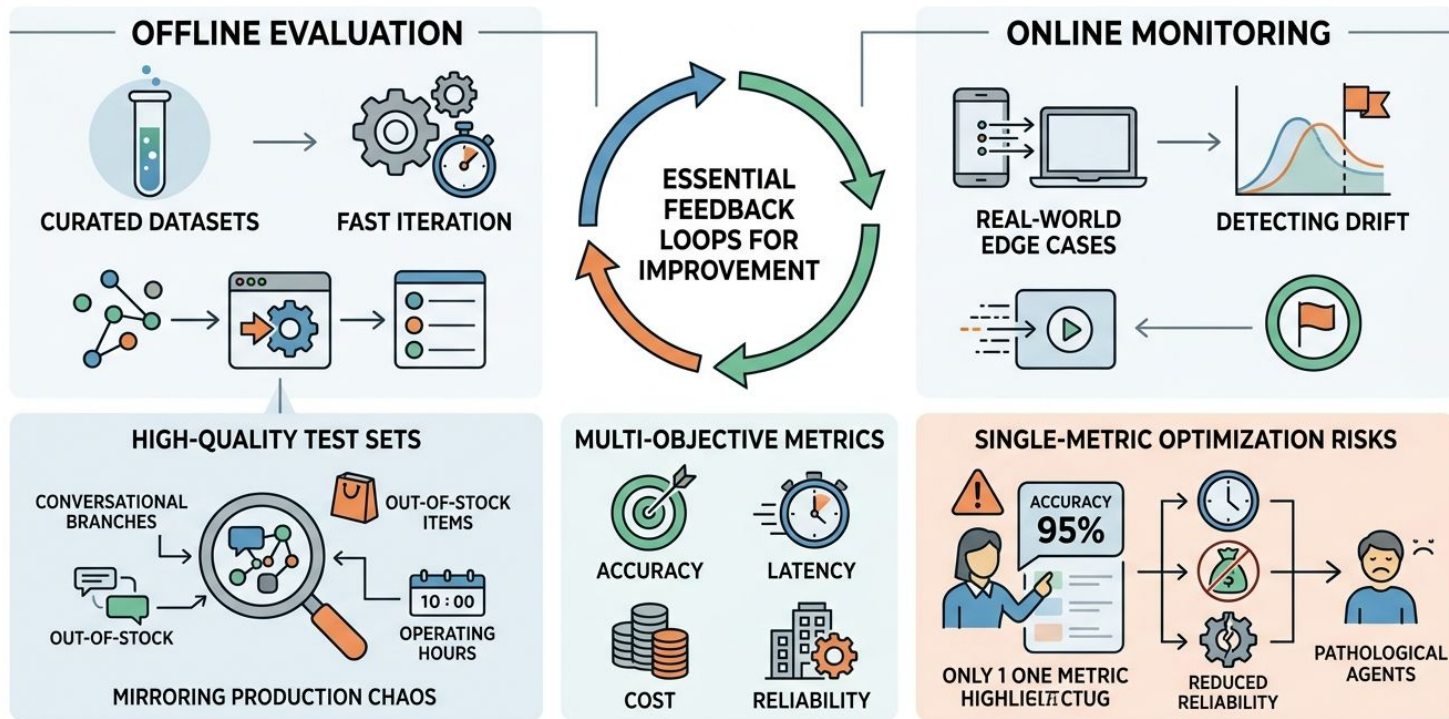
U.S. General Services
Administration



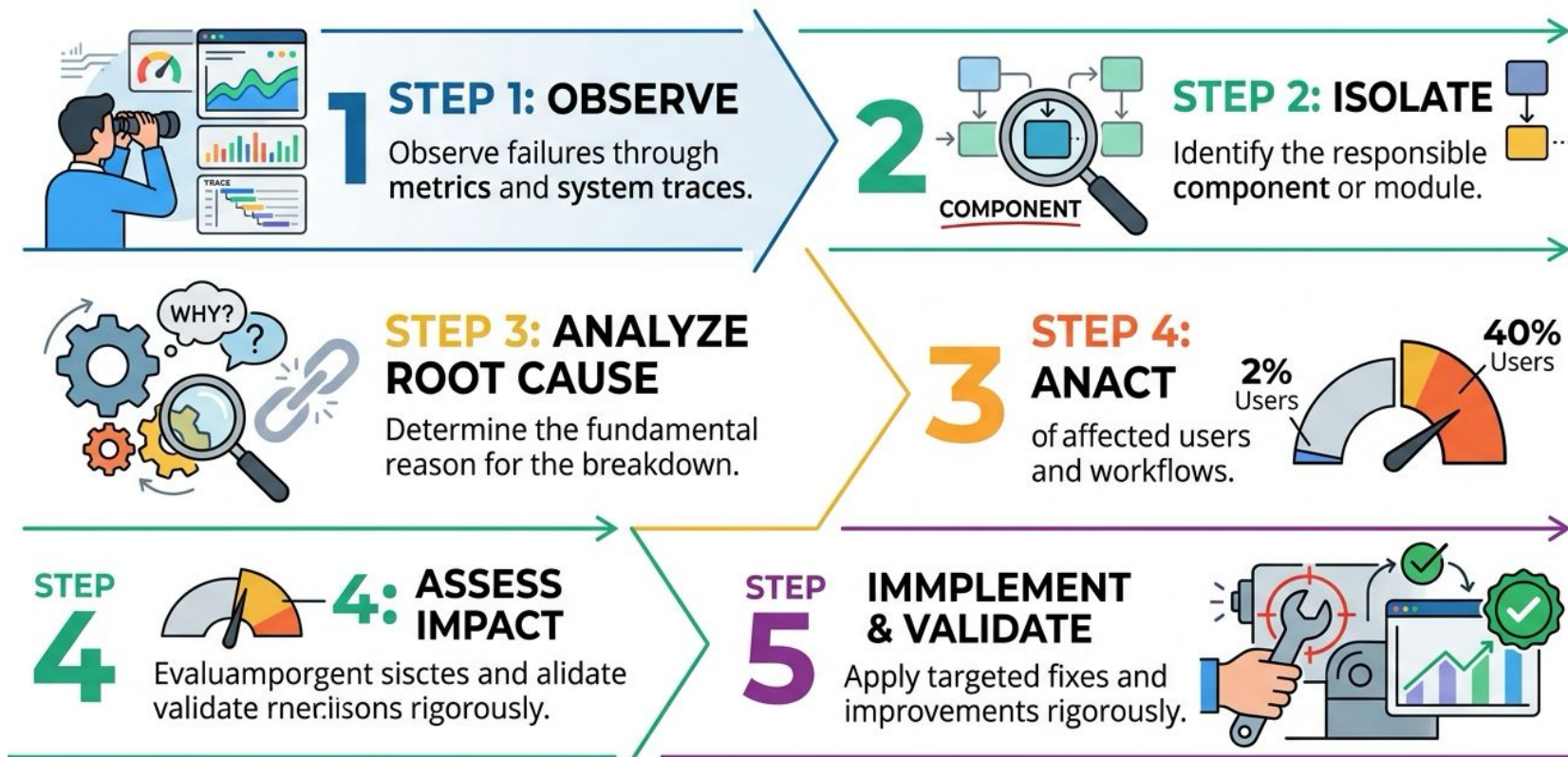
Chapter 3.5: Prompt Optimization, Few-Shot Learning, and Fine-Tuning

US General Service Administration
AI Community of Practice - March 2026

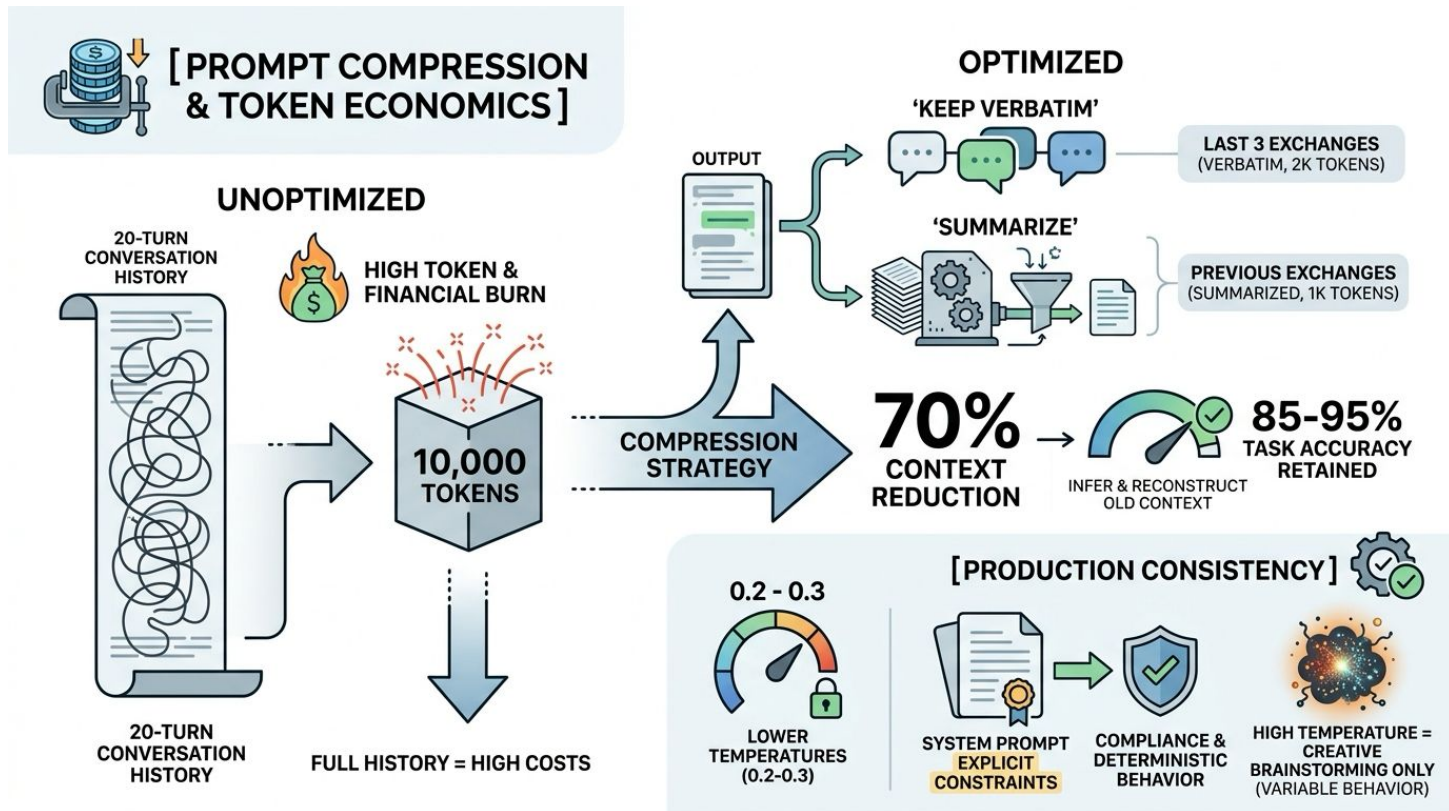
Evaluation Pipelines as Optimization Foundations



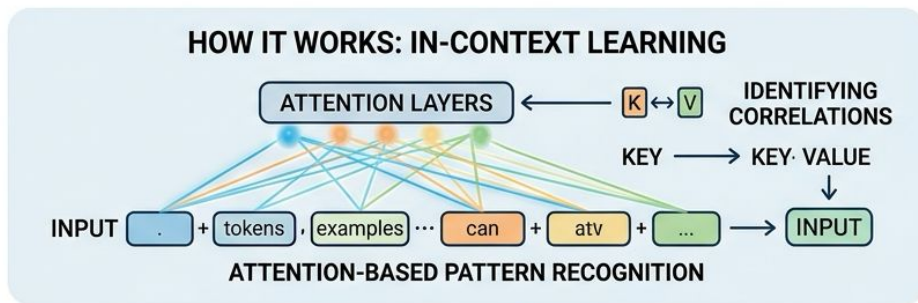
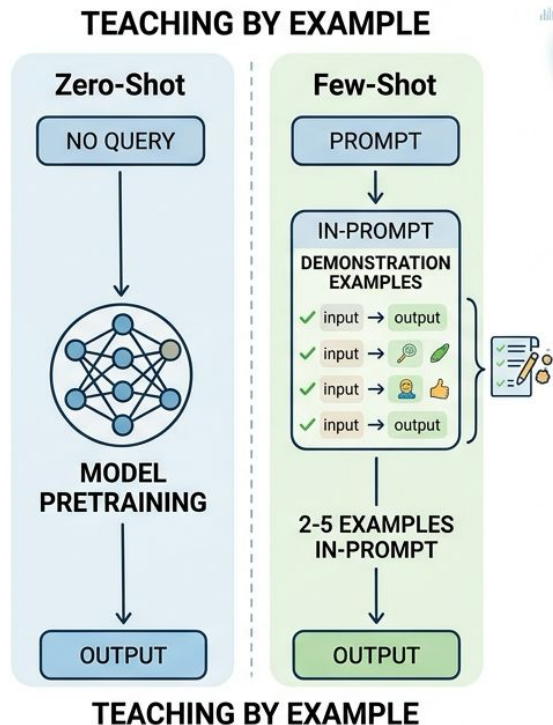
Systematic Optimization -- The Five-Step Diagnostic



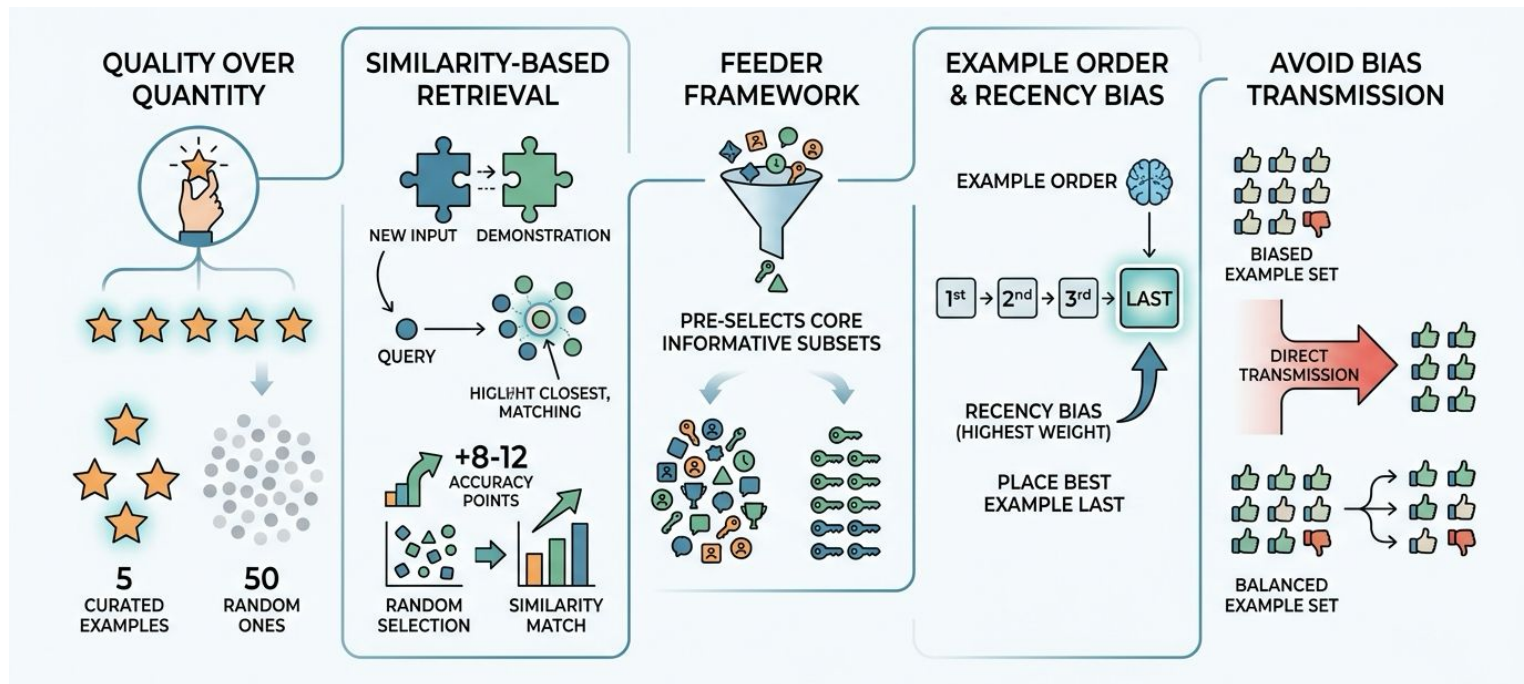
Prompt Compression and Token Economics



Few-Shot Learning -- Teaching by Example



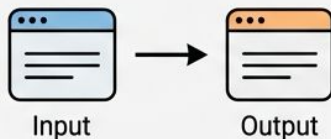
Demonstration Quality and Selection Strategies



Chain-of-Thought Prompting for Reasoning

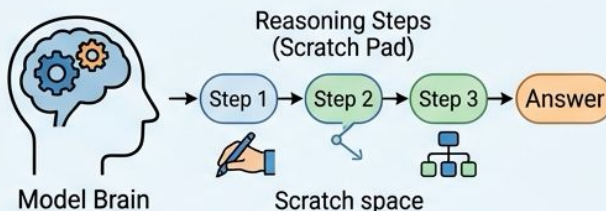
STANDARD PROMPTING

Only Input-Output Examples

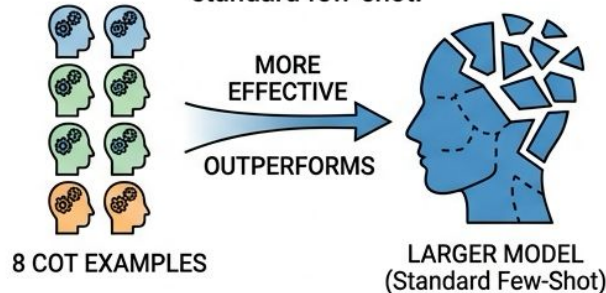


CHAIN-OF-THOUGHT PROMPTING

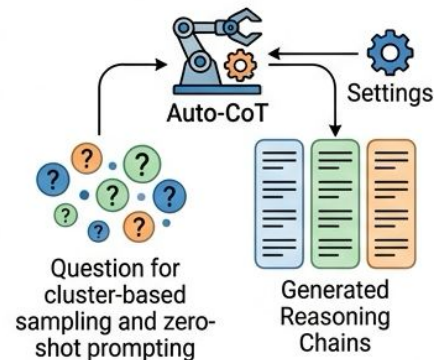
Shows Intermediate Reasoning Steps



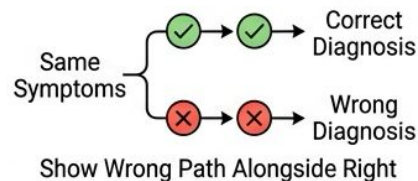
8 CoT examples beat larger models using standard few-shot.



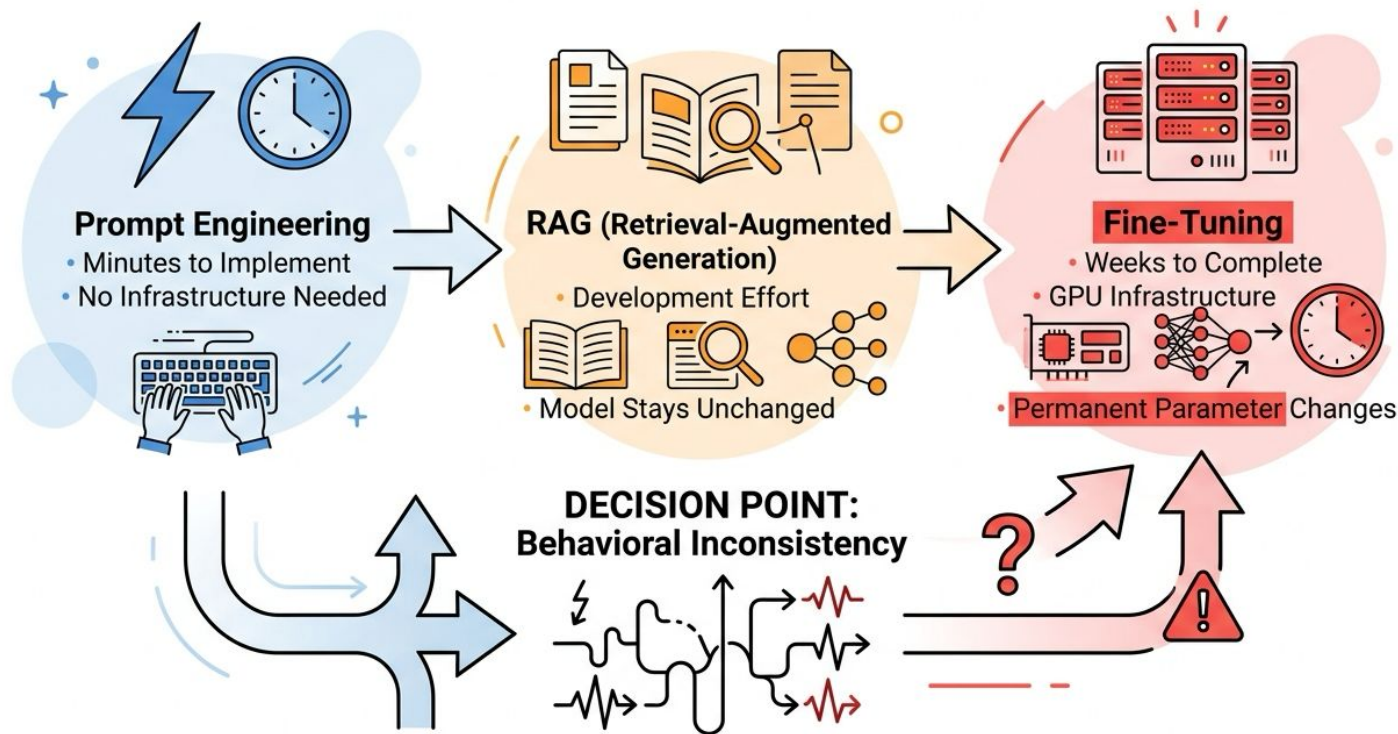
AUTO-COT: AUTOMATED REASONING CHAINS



CONTRASTIVE COT: Learning Right and Wrong



When to Escalate from Prompting to Fine-Tuning



Learning from Agent Trajectories

SUCCESSFUL TRAJECTORIES



TEACH OPTIMAL
DECISION PATTERNS

FAILURE TRAJECTORIES



TEACH WHAT
TO AVOID

AGENTBANK



50,000+
TRAJECTORIES



40-50%
IMPROVEMENT
OVER BASE

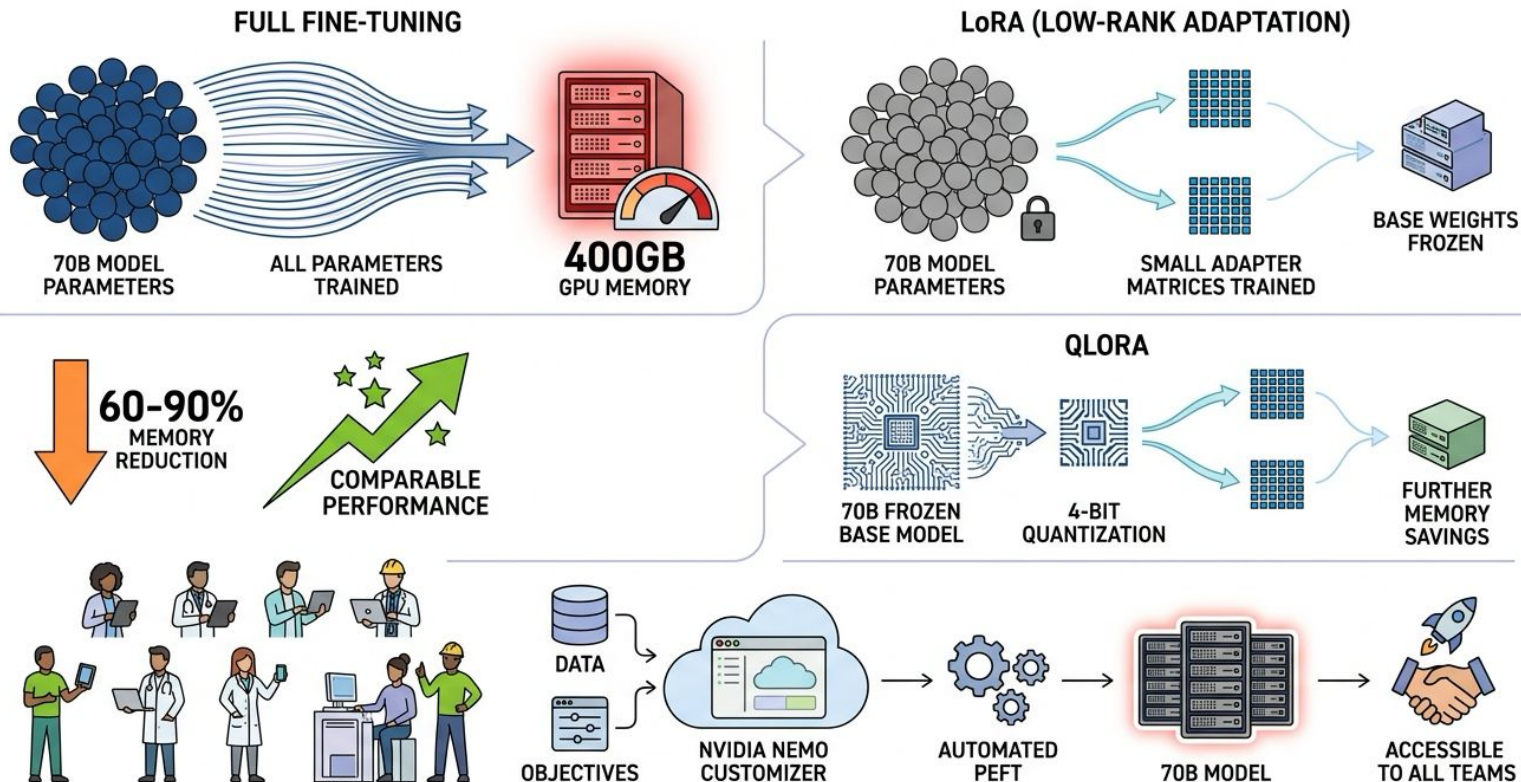


QUALITY FILTERING
via
REJECTION FINE-TUNING



REMOVES
NOISE

Parameter-Efficient Fine-Tuning with LoRA



RLHF -- Aligning Agents to Human Preferences

- Some behaviors resist explicit demonstration
 - RLHF learns from comparative human judgments
 - Stage 1: SFT baseline; Stage 2: reward model; Stage 3: RL
 - DPO simplifies to two stages, reducing compute
 - Beware annotation biases: length, confidence, fluency
- 

Key Takeaways

- Prompt optimization is engineering, not art -- measure everything
 - Few-shot examples: quality and order beat quantity
 - Fine-tune only after prompts and RAG reach their ceiling
 - PEFT democratizes fine-tuning; RLHF captures preferences
 - Next: trace analysis reveals why agents fail step-by-step
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